

Quintin Hypothesis Benchmark v1

First Real Run: Results and Statistical Analysis

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Abstract

We report the first end-to-end run of the Quintin Hypothesis (QH) benchmark v1 against the firm’s contradiction-geometry probe and two baselines (a random runner and a cosine-only runner). The frozen v1 dataset contains 1936 items spanning 3 domains. All runners use the deterministic hash-det-v1 embedder, so the comparison is on geometry, not on embedding quality.

Honest negative finding. On this run a non-firm baseline beats the firm’s contradiction-geometry probe on at least one slice. The benchmark was built so the firm could be wrong in public; the losing slices are enumerated in Section 3, not buried. The MQS-on-the-firm-probe composite is below the announcement threshold, so this result is published but not promoted.

1 Run envelope

Run stamp	20260514T052854Z
Benchmark version	qh-v1
Git SHA	0034929158a42e4e536d85efd41ab22721c7ca50
Git branch	main (dirty=True)
Dataset sha256	b25ab62102389fbbbf2cdad08cc5a056e4946631835a89672fe88aa0a8fbe7c4
Dataset items	1936
Dataset frozen verified	True (passed)
Embedder	hash-det-v1 (dim 192)
Random-runner seed	0
Analysis bootstrap seed	17
Bootstrap	10000 resamples, paired BCa (bias-corrected and accelerated)
Embedding budget	0 / 50000 credits
Python	3.13.7

2 Leaderboard

Runner	n of N	Accuracy	AUROC	ECE	Latency p50 (ms)
contradiction_geometry	1936 of 1936	0.2877	0.5858	0.2752	0.0065
random	1936 of 1936	0.3352	0.4964	0.2537	0.0022
cosine	1936 of 1936	0.3673	0.3987	0.3964	0.0035

3 Honest findings

A non-firm baseline wins on the following slices:

- Overall 3-way accuracy: the cosine baseline (0.3673) beats the firm’s contradiction-geometry probe (0.2877).
- Overall 3-way accuracy: even the random runner (0.3352) beats the firm probe (0.2877) — the probe is below chance on accuracy.
- Domain ‘economics’ accuracy: cosine (0.4094) beats the firm probe (0.2596).
- Domain ‘ethics’ AUROC: cosine (0.5124) beats the firm probe (0.4667).
- Domain ‘physics’ accuracy: cosine (0.4037) beats the firm probe (0.2898).
- The firm probe is degenerate on this run: it predicts ‘contradicting’ for every item. Its AUROC is not meaningless, but its accuracy reflects only the base rate of that label.

4 Statistical analysis: firm probe vs cosine

Paired comparison over 1936 aligned items. Confidence intervals are paired BCa bootstrap intervals (10000 resamples); positive values favour the firm probe.

4.1 3-way accuracy difference

Δ accuracy (firm – cosine)	-0.0795
95% BCa CI	[-0.1152, -0.0429]
Excludes zero	True
Bootstrap two-sided p	0.0001
Effect size (Cohen’s h)	-0.1698 (negligible)
BCa internals	$z_0=0.0015$, $a=0.0005$

4.2 McNemar’s test (paired 3-way correctness)

Method	mcnemar-chi2-continuity
b (firm right, cosine wrong)	553
c (firm wrong, cosine right)	707
Discordant pairs	1260
Statistic	18.5786
p -value	0.0000
Odds ratio (b/c)	0.7822

4.3 AUROC difference (contradicting vs coherent)

AUROC firm	0.5858
AUROC cosine	0.3987
Δ AUROC (firm – cosine)	0.1871
95% BCa CI	[0.1316, 0.2424]
Excludes zero	True
Bootstrap two-sided p	0.0001

4.4 Per-domain accuracy difference (firm – cosine)

Domain	n pairs	Δ accuracy	95% BCa CI	Cohen’s h
economics	574	-0.1498	[-0.2160, -0.0854]	-0.3194 (small)
ethics	344	0.1395	[0.0640, 0.2093]	0.3213 (small)
physics	1018	-0.1139	[-0.1660, -0.0658]	-0.2402 (small)

5 MQS-on-the-firm-probe (announcement gate)

Composite quality score for the contradiction-geometry probe: **0.4741** (threshold 0.50). The composite is below the threshold, so the announcement tweet is suppressed. The result is published but not promoted.

Component	Value
accuracy_lift	0.0000
auroc_lift	0.1715
calibration	0.7248
beats_cosine	1.0000

6 Reproducibility

Every number in this document is produced by `noosphere.benchmarks.qh_analysis` from the frozen dataset and the recorded envelope; no value is hand-edited. To reproduce: check out git SHA 0034929158a42e4e536d85efd41ab22721c7ca50, confirm the dataset sha256 matches the envelope, and re-run `noosphere/scripts/run_qh_full.sh`. The random runner is reproducible from the recorded seed; the cosine and firm probes are deterministic.